

# ml.assignment 3

## Abstract

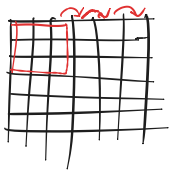
On the given image, apply filters

1. Mean filter 3x3
2. Median filter 3x3
3. Gaussian 5x5 with  $\sigma = 2$
4. sharpening filter

Save the img aafter each type of filter along with original image in a pdf file.

## workout

1) Mean filter



$$\begin{aligned} \text{Output dim} &= \left( \frac{l-f}{\text{stride}} \right) + 1 \\ &= \frac{307-3}{1} + 1 \\ &= \underline{\underline{305}} \end{aligned}$$

## Mean filter

Output



## Mean filter

```

pixels = 1d img
stride = 1
filter size = 3x3
output dimension = img_length - (filter size/stride) + 1
for i in range(out_dim):
    for j in range(out_dim):

```

Filter Logic

$l = \text{output dim}$

for  $r$  in range( $l$ ):

for  $c$  in range( $l$ ):

start\_y =  $r \times \text{stride}$

start\_x =  $c \times \text{stride}$

# Here we do the filtering

for  $ky$  in range(filter\_h):

*filter  
applying here*

for  $kx$  in range(filter\_w):

val = pixels[i][j]

*filter*

↳ start point + (3px)

current\_sum += val

*change this  
for other  
filters.*

*calculates the sum of all 9 pixels in the filter*

mean = current\_sum / filter area

for each filter apply → get one output pixel from mean

Completed  
output  
image ←  $r^{\text{th}}$  col &  
 $j^{\text{th}}$  row  
of output\_img

**Median Filter**

output



```

nums = []
for ky in range(f_size):
    for kx in range(f_size):
        i = {Pix coords
        j = }
        val = pixel[i][j]
        nums.append(val)
nums_sorted = nums.sort()
if len(nums_sorted) % 2 == 0:
    median_even
if len(...) % 2 == 1:
    median_odd

```

---

## Gaussian Filter

output



## Gaussian Filter

2-D zero-mean gaussian fn

$i, j \rightarrow$  dist from pixel at the centre

$$g(i, j) = e^{-\frac{i^2 + j^2}{2\sigma^2}}$$

Filter size =  $5 \times 5$

→ Separable,  $\sim e^{-\frac{i^2}{2\sigma^2}} \times e^{-\frac{j^2}{2\sigma^2}}$

1) create gaussian filter

```
radius = size//2
```

Since its zero mean, the filter center is our reference

for x from - radius to + radius:

for y from -radius to + radius:

```
power = -(x^2+y^2)/2*sigma^2
```

```
value = e^(power)
```

```
[x + radius][y+radius] = value
```

Iteration logic for  $3 \times 3$  filter

$$x = -1 \quad y = -1 \text{ (Image)}$$

$$\text{row} = -1 + 1 = 0, \text{col} = -1 + 1 = 0$$

$$\text{exp} = - \frac{(-1)^2 + (-1)^2}{2} = -\frac{2}{2} = -1$$

$$\text{Value} = e^{-1} = 0.3676$$

$$\therefore \text{filter}[0][0] = 0.3676$$

Top middle

$$x = -1 \quad y = 0$$

$$\text{row} = 0, \text{col} = 1$$

$$\text{exp} = \frac{(-1)^2 + 0^2}{2} = -0.5$$

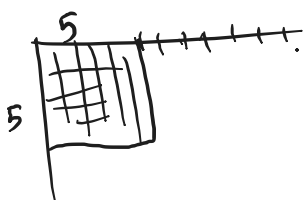
$$\therefore \text{filter}[0][1] = 0.6065$$

The kernel matrix would look like this

$$\begin{bmatrix} 0.368 & 0.607 & 0.368 \\ 0.607 & 1.000 & 0.607 \\ 0.368 & 0.607 & 0.368 \end{bmatrix}$$

Gaussian - Filter

$$\text{filter} = 5 \times 5$$



$$f(x) = \left( \frac{1}{\sigma\sqrt{2\pi}} \right) e^{-\frac{i^2 + j^2}{2\sigma^2}}$$

for  $x$  in range  $(-r, r+1)$ .



for  $y$  in range  $(-r, r+1)$

Apply this  $\left[ e^{-\frac{i^2+j^2}{2\sigma^2}} \right]$  to get  
gaussian matrix

$$D_0 \left( \frac{1}{2\pi\sigma^2} \right) G[ ] \rightarrow$$

Normalized const  
sum of all gaussian  
weights = 1

Normalized const

$$1D : \frac{1}{\sqrt{2\pi} \cdot \sigma}$$

$$2D : \frac{1}{2\pi \cdot \sigma^2}$$

→ This is applicable during integration S-T sum = 1  
→ But we will divide by sum for normalizing

## Sharpen Filter

output



## Sharpen filter

Basically in this case, filter output is the average after scaling the pixel frame using the matrix

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$